A Tiny Motor Inspired by an Ancient Screw

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Scientists have built a remarkable **molecular** motor that spins when electricity passes through it and reverses direction when the **voltage** is switched.

Nature already has **molecular** motors inside living cells, and researchers have spent decades trying to copy them, work recognized with the 2016 Nobel Prize in Chemistry. The first electricity-powered **molecular** motors, built in 2011, sat on surfaces and relied on a scanning tunneling microscope tip to drive them.

The new motor borrows its design from the Archimedes screw, one of the oldest machines in recorded history. Traditionally used to move water uphill, it uses a rotating spiral to transport material. Here, electrons travel through a spiral-shaped molecule, and their flow makes the molecule rotate.

The spiral molecule, created by Prof. Michael Pittelkow's group at the University of Copenhagen, measures only a few millionths of a millimeter and attaches to gold through sulfur atoms. At Israel's Weizmann Institute, Prof. Oren Tal and Dr. Anil Kumar Singh placed molecules between two atomically sharpened gold tips, stretching or compressing the screw by changing the gap.

When **voltage** was applied, the molecule rotated, and reversing the **voltage**'s polarity reversed its direction: the first current-driven **molecular** motor firmly connected to two metal **electrodes** with controllable direction. Electrons transfer **angular momentum** to the spiral as they pass, like moving air turning a wind turbine, and above a **voltage** threshold rotation becomes continuous.

Rotation brought a surprise: the spinning molecule conducted electricity far better than when still. Simulations by Prof. Fabian Pauly and PhD students Léon Martin and Yuta Ito at the University of Augsburg, run on the LiCCA supercomputer, showed **conductivity** rising by as much as 100 times during rotation. The positions that conduct best are ones the molecule naturally avoids at rest, Martin explained: only rotation makes them accessible.

Word Treasure

molecular -- About single molecules, the tiny building blocks of matter.

voltage -- The electrical push that drives a current.

electrodes -- Metal ends that carry electricity into or out of something.

conductivity -- How well something carries electricity.

angular momentum -- The spinning push carried by moving things.

Background you need

A molecule is a tiny group of atoms joined together, far too small to see.

An electric current is a flow of electrons through a material.

The Archimedes screw is an ancient device that lifts water with a turning spiral.

Gold is a metal that conducts electricity well, which is why it is used for tiny electrical contacts.

Break it down

LEAD: A tiny motor inspired by an ancient screw
KEY EVENT: A current-driven molecular motor is built
+- WHO: Pittelkow's group, Tal and Singh, and Pauly's team
+- HOW: Electrons flow through a spiral molecule; voltage reverses its direction
+- FIRST: Firmly connected to two electrodes with controllable direction
SURPRISE: Conductivity rises as much as 100 times during rotation
QUOTE: Martin says only rotation makes the best-conducting positions accessible

Why it matters

Machines this small could one day power tiny devices, and this motor shows how an ancient idea can solve a modern engineering puzzle.

Test what you remember

1. What ancient device inspired the new motor?

(A) Scanning tunneling microscope
(B) Wind turbine
(C) Archimedes screw
(D) LiCCA supercomputer

2. What happens when the voltage's polarity is reversed?

(A) It breaks apart
(B) It conducts 100 times better
(C) It stops
(D) It reverses direction

3. Who created the spiral molecule?

(A) Prof. Oren Tal
(B) Prof. Fabian Pauly
(C) Dr. Anil Kumar Singh
(D) Prof. Michael Pittelkow's group

4. By how much did conductivity rise during rotation?

(A) Two times
(B) A few millionths of a millimeter
(C) As much as 100 times
(D) Ten times

5. On what were the simulations run?

(A) University of Augsburg
(B) University of Copenhagen
(C) Weizmann Institute
(D) LiCCA supercomputer

6. What did Martin say about the best-conducting positions?

(A) They are made of gold
(B) They only appear above a voltage threshold
(C) They are positions the molecule prefers at rest
(D) They are positions the molecule naturally avoids when at rest

Answer key (try the quiz first!): 1: C 2: D 3: D 4: C 5: D 6: D

Questions to think about

1. Léon Martin: Only rotation makes the best-conducting positions accessible.

MY THOUGHTS (at least 20 words):